BrainLM: Toward a Foundational Model of Cognition Bridging Brain Imaging and Language

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Introduction

BrainLM is a multimodal foundation model designed to analyze and interpret large-scale brain imaging data. Pretrained on 6,700 hours of fMRI data, it predicts clinical variables such as age, anxiety, and PTSD, and generalizes to unseen cohorts. By integrating behavioral and clinical metadata using a language model interface, BrainLM enhances interpretability and supports intuitive, text-based analysis of brain function—paving the way for more accessible and theory-driven neuroscience.

Pre-trained fMRI Encoder

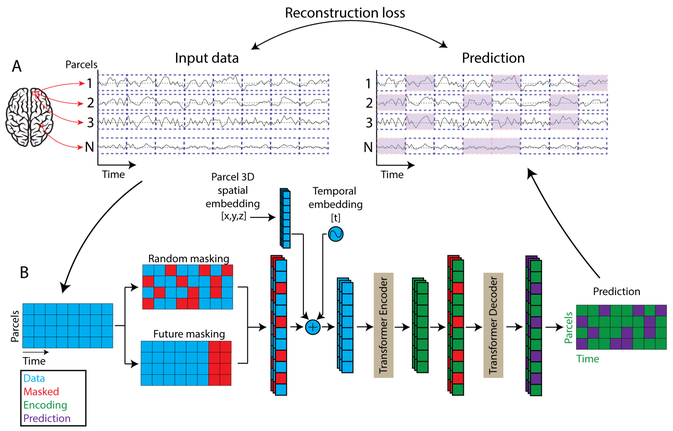


Figure 1. fMRI encoder architecture and pre-training using reconstruction-based loss

Table 1. Results on downstream regression of clinical variables [1].

	Age	PTSD (PCL-5)	Anxiety (GAD-7)	Neuroticism
Raw data	2.0 ± 0.2219	0.034 ± 0.0027	0.172 ± 0.0066	0.160 ± 0.0137
SVR	0.659 ± 0.036	0.022 ± 0.004	0.090 ± 0.010	0.087 ± 0.008
MLP	0.693 ± 0.001	0.021 ± 0.0003	0.081 ± 0.001	0.079 ± 0.0005
LSTM	0.596 ± 0.040	0.019 ± 0.001	0.083 ± 0.0022	0.076 ± 0.002
GCN	0.862 ± 0.09	0.021 ± 0.002	0.083 ± 0.02	0.077 ± 0.006
BrainLM	**0.485 ± 0.0252**	**0.018 ± 0.0008**	**0.074 ± 0.0053**	**0.072 ± 0.0049**

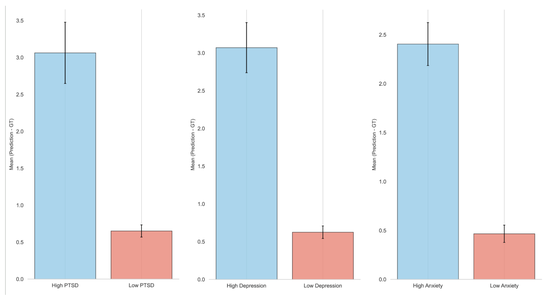


Figure 2. BrainLM reveals higher "brain age" for patients with psychiatric conditions.

Vision Language Models (VLM) Training

Goal: Integrate the BrainLM vision encoder with pre-trained multimodal language models to encode behavior and clinical metadata in a textual interface

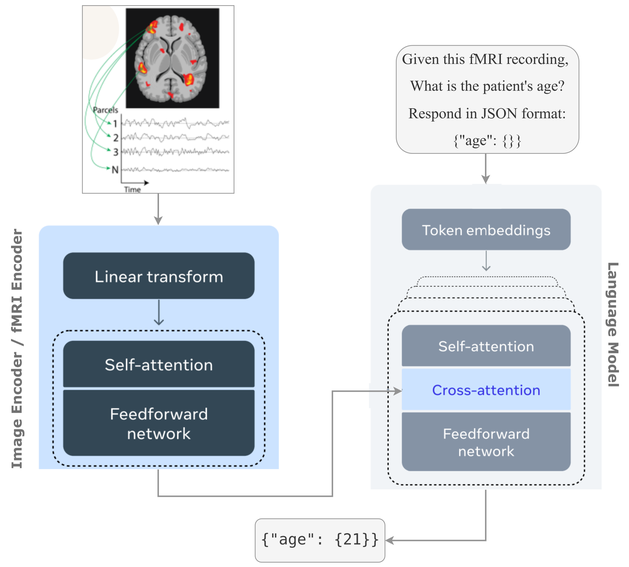


Figure 3. Multimodal language model architecture [2].

- Evaluated two Vision-Language Models: **LLaMA 3.2 11B** and **Qwen 7B**
- Compared performance on age prediction using native visual encoder vs pre-trained BrainLM encoder paired with **GPT-2**

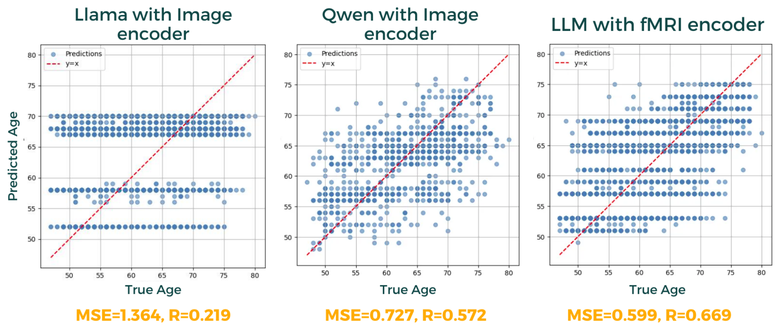


Figure 4. Comparative results of image and fMRI encoder.

Key findings: Using a specialized fMRI encoder outperforms VLM visual encoders, reflecting the complexity of fMRI data and the distribution shift from the VLM pre-training

Adversarial Training for Text-fMRI Alignment

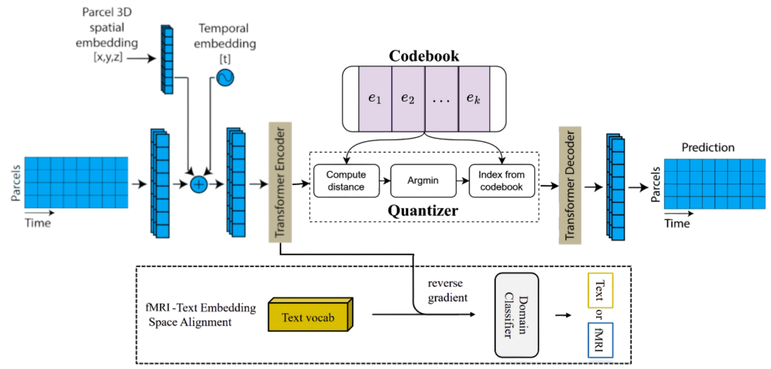


Figure 5. Adversarial Vector Quantized Autoencoder architecture.

The architecture consists of an **Autoencoder** for learning complex representations of fMRI, a **Vector Quantizer** for discretizing them and creating a vocabulary, and a **Domain Classifier** for modality alignment [3].

Self-Supervised Pre-Training

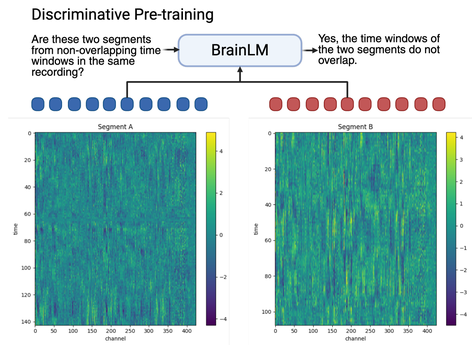


Figure 6. Self-supervised pre-training using a discriminative objective with augmented data

In absence of ground-truth text-aligned fMRI data, we adapt the approach from [4] to predict synthetic data augmentations.

References

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